Lab 1 – Smart City

Nick Kooshki

Old Dominion University

CS 411W

Dr. Sumaya Sanober

27 February 2026

Version 1

Table of Contents

1 Introduction..... 3

2 Product Description..... 3

3 Identification of Case Study..... 7

4 Glossary..... 8

5 References..... 9

List of Figures

Figure 1: Smart City Major Functional Component Diagram.....5

List of Tables

Table 1: Comparison Between RWP and Prototype..... 6

1 Introduction

Cities are getting bigger and busier every year. The United Nations says that by 2050, around 6.5 billion people will be living in cities. Because of this, people deal with more traffic, less space, and slower communication between city systems. The way most cities are built cannot handle this kind of growth, so things like traffic, pollution, and poor resource management keep getting worse. Smart City is our solution to this problem. It is a community-based app that helps people get around town faster and stay informed. The app uses AI and real-time data from users to make smarter route suggestions and share useful updates. The main goal is to reduce congestion and make it easier for residents, commuters, and city officials to stay connected. Smart City helps people and technology work together so cities can be more organized, efficient, and enjoyable to live in.

2 Product Description

Smart City is an app that gives people live updates about what is happening around them. It shows real-time traffic, crowd levels, and route suggestions based on data from users and sensors. The app does not just guide users; it learns and updates itself to make sure everyone gets where they need to go without the stress of congestion. One of the main features that makes Smart City different is its Alternative Destination system. If too many people are going to the same place, the app uses live data to suggest other nearby spots that are not as crowded. It connects with the Google Maps API and works with a dynamic queue system that spreads users evenly. This helps fix the Braess Paradox, which is when rerouting people accidentally creates new traffic jams.

Key product features include an engaging user interface with a community dashboard and navigation capabilities. The system ensures efficient data handling by processing and storing

collected information to provide AI-assisted insights. Smart City is significant and unique because it aims to eliminate barriers to data collection while providing a one-stop solution for citizens to navigate their communities simply. It specifically addresses the global problem of rapid urbanization and overcrowding. By implementing this software-based solution, the goal is to give citizens the tools they need to navigate their community without hassle. The application gathers user data in a secure and private manner, which city officials and planners can then analyze to implement changes regarding roadways and highways.

The hardware required to support this solution includes a computer to access the web application for data handling, and a smartphone (iOS or Android) for users to access the navigation and community hub. Internet connectivity is essential for engagement. Structurally, the solution is divided into three layers **(see Figure 1)**. The presentation layer allows users and city officials to engage with the application. The application layer provides the logical components, including core data handling logic. Finally, the data layer houses the database used for storage. The user application and data platform modules will be developed using the React Native Framework, interacting with APIs like the Google Maps Platform. JavaScript will be used for the frontend and Java for backend, using the Firebase library for the database. To outline the specific scope of this project, a feature comparison between the fully envisioned Real-World Product (RWP) and the current prototype is provided below **(see Table 1)**.

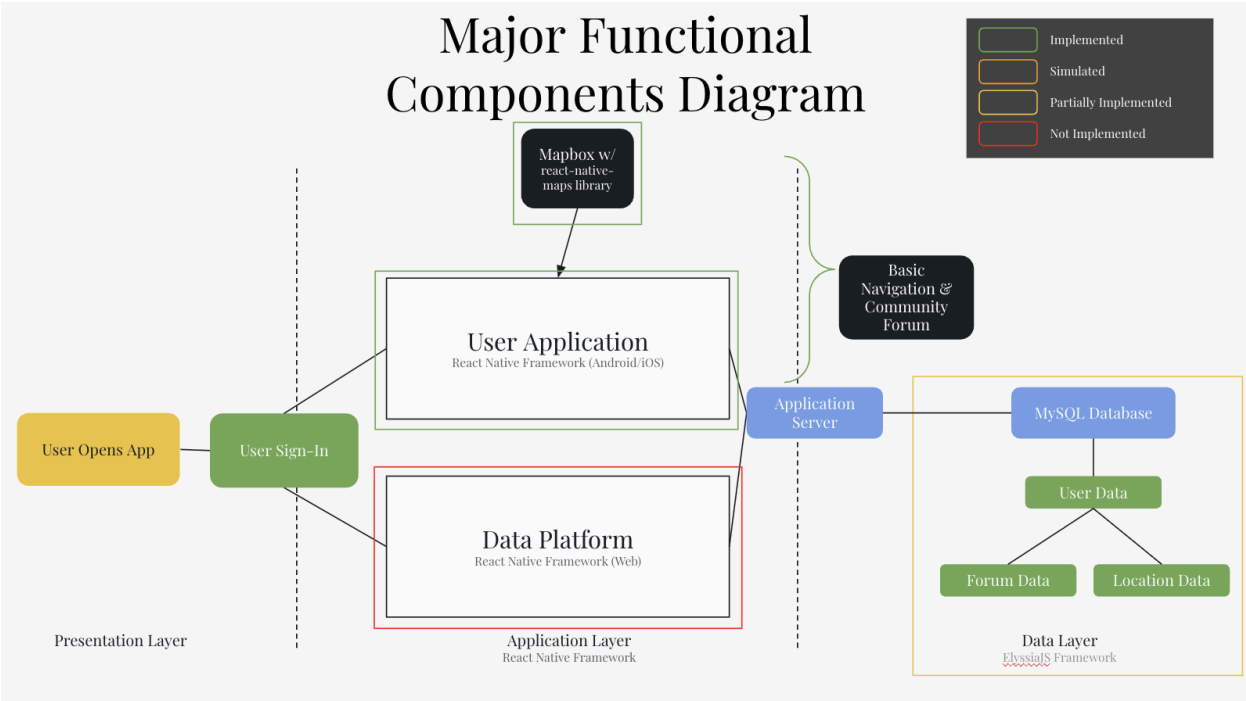

Figure 1. Smart City Major Functional Component Diagram.

Table 1. Comparison Between RWP and Prototype.

Feature Table: RWP vs. Prototype

Category	Features	RWP	Current Prototype	Reasoning
Account Management	Login/Authentication	Fully Implemented	Fully Implemented	Essential for prototype
	Account Creation/Deletion	Fully Implemented	Fully Implemented	Essential for prototype
	Manage Account Access/Permissions	Fully Implemented	Fully Implemented	Essential for prototype
User Application	Access Community Forum	Fully Implemented	Fully Implemented	Essential for prototype
	Access Dynamic Queue System for Navigation	Fully Implemented	Partially Implemented	We removed the Dynamic Queue implementation as it would be too complex for the given time frame. Still implementing the basic navigation.
	Alerts and Notifications	Fully Implemented	Partially Implemented	Simulated feature until we figure out how to handle API calls.
	Add Parking locations	Fully Implemented	Not Implemented	No time given scope of project. Could reconsider

Feature Table: RWP vs. Prototype Continued

Category	Features	RWP	Current Prototype	Reasoning
Data Platform	Central Information Hub	Fully Implemented	Partially Implemented	Still will implement the database for storing information in. Will drop long-term data platform due to time constraints.
	Data Analytics and Insights	Fully Implemented	Not Implemented	Dropped due to no longer implementing the long-term data platform
	Visualize Data	Fully Implemented	Not Implemented	Dropped due to no longer implementing the long-term data platform
	Data Anonymity	Fully Implemented	Fully Implemented	Essential for prototype

3 Identification of Case Study

This product is being developed for local city residents and commuters who desire real-time traffic, event, and safety updates, as well as community members who wish to share local information. It also serves city officials who require data insights to improve decision-making regarding infrastructure and resource management. The application is designed to make commuting easier, reduce traffic issues, and help people stay informed by combining navigation, community input, and AI into a single platform.

The initial case study group will consist of a small group of Old Dominion University (ODU) students (5–10) and nearby residents who commute around the campus and surrounding area. The university campus environment serves as an ideal microcosm of a smart city due to its mix of pedestrian, vehicle, and shuttle traffic, particularly during peak class change times. Participants will use the prototype to validate the accuracy of crowd levels and route suggestions in this high-density environment. Feedback will specifically focus on the responsiveness of the "Alternative Destination" system and the clarity of the user interface during active navigation.

Future uses could expand to the broader city populations of Norfolk and Hampton Roads, allowing for larger-scale data collection and analysis. Additionally, local businesses could utilize the platform to share events, while visitors or tourists could use the app for navigation and discovering local updates.

4 Glossary

Access Control: controls that enforce authorized access to information and system resources in line with policy

Anomaly Detection: identifying data patterns that deviate from expected behavior

Application Programming Interface (API): a set of features and rules that lets software interact with other software (a "contract" between programs)

Citizen Feedback Loop: a two-way process where institutions act on and report back about citizen input to sustain engagement and trust

Crowdsourced Reporting: gathering information by soliciting contributions from a large group of people, typically via the internet

Internet of Things (IoT): systems that combine sensing, computing, communication, and actuation to connect physical/cyber objects

Open Data (Open Data Access): data anyone can freely use, re-use, and redistribute, with at most attribution/share-alike requirements

Predictive AI / AI-Powered Forecasting: using data, statistical algorithms, and ML to estimate the likelihood of future outcomes

Privacy-by-Design ("Privacy-First Design"): building systems so that, by design and by default, only the personal data necessary for a purpose are processed

5 References

Gorzelany, J. (2024, August 30). Here's where commuters lose the most time and money sitting in traffic, study shows. Forbes. https://www.forbes.com/sites/jimgorzelany/2024/08/30/heres-where-commuters-lose-the-most-time-and-money-sitting-in-traffic-study-shows/

Joyce, H. (2025, March 26). Urban overcrowding: A major driver of pollution crises. ShunWaste. https://shunwaste.com/article/how-is-overcrowding-in-cities-causing-more-pollution

King, G. (n.d.). Data sharing: The next step in municipal collaboration. Data-Command. https://www.data-command.com/the-cloud/data-sharing-the-next-step-in-municipal-collaboration

Lee, S. (2025, May 27). Understanding urban overcrowding: The social and economic implications of overcrowding in cities. Number Analytics. https://www.numberanalytics.com/blog/ultimate-guide-to-overcrowding-in-urban-sociology

Luoma, K. (2016, November 21). How does population growth impact rapid urbanization? Population Education. https://populationeducation.org/how-does-population-growth-impact-rapid-urbanization/

Poindexter, D. (2018, January 24). The effects of overcrowding and the behavioral sink theory. The Borgen Project. https://borgenproject.org/theory-behavioral-sink/

Smith, H. (2021, June 18). Post-COVID L.A. traffic analysis: Has rush hour changed? Los Angeles Times. https://www.latimes.com/california/story/2021-06-18/post-covid-l-a-traffic-analysis-has-rush-hour-changed

Weinberg, N. (2016, August 11). How can cities be preemptive and effective in preventing overcrowding? Data-Smart City Solutions.

https://datasmart.hks.harvard.edu/news/article/how-can-cities-be-preemptive-and-effective-in-preventing-overcrowding-888

Zurich Insurance Group. (2023, April 13). The risks of rapid urbanization in developing countries. https://www.zurich.com/knowledge/topics/global-risks/the-risks-of-rapid-urbanization-in-developing-countries